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RDL-YOLO: an efficient algorithm for detecting road surface defects

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Road surface defect detection is essential for intelligent transportation, autonomous driving safety, and preventive road maintenance. However, accurately identifying potholes and protrusions under complex illumination, irregular morphology, and multi-scale variations remains challenging, especially for lightweight models deployed on edge devices. This paper proposes RDL-YOLO, an efficient and lightweight detection framework built upon the YOLO11n architecture. The model integrates the C3k2_RVB and C3k2_REVB modules to enhance multi-scale feature extraction and improve robustness under low-light conditions. A shared detection head LSCD_LQE is introduced to jointly optimize classification and localization accuracy by incorporating learnable quality estimation. The DySample module enhances adaptive feature upsampling and spatial detail restoration, while DWConv reduces computational overhead to achieve real-time inference on resource-constrained platforms. Experimental results on a road-defect dataset demonstrate that RDL-YOLO achieves notable performance gains. Precision increases by 0.63 percent, recall increases by 4.24 percent, the F1-score improves by 1.82 percent, and mAP at 0.5 improves by 1.76 percent compared with the baseline YOLO11n model. Meanwhile, computational complexity is significantly reduced, with GFLOPs decreasing by 44 percent and parameter count decreasing by 25 percent. Cross-dataset generalization tests further verify the robustness and adaptability of the proposed model in diverse road conditions. The results confirm that RDL-YOLO provides an effective and lightweight solution for real-time road defect detection and intelligent vehicle hazard avoidance.

Keywords Road defect detection, YOLO11, Lightweight detection, Multi-scale feature extraction, Edge computing

As modernization progresses and traffic volume increases, road faults have grown more pronounced. These concerns not only undermine driving comfort and expedite vehicle deterioration but also have the potential to induce traffic accidents, so posing a significant risk to travel safety[1–3]. Consequently, the efficient and timely identification of pavement faults is essential for maintaining road traffic safety and protecting human lives and property[4,5].

Potholes and elevated obstructions are among the most common types of road infrastructure issues. These faults compromise pavement structural integrity and can lead to loss of vehicle control, vehicle damage, and traffic accidents, hence posing significant risks to driving safety[6,7]. Consequently, attaining efficient and accurate pavement problem identification alongside prompt evasive measures has emerged as a crucial concern for improving road safety guarantee[8,9]. Conventional approaches dependent on manual inspections and subjective avoidance need considerable human resources and are plagued by inefficiency, inconsistent results, and safety risks. These methodologies fail to satisfy the requirements of contemporary intelligent transportation systems for elevated reliability, real-time performance, and automation levels[10].

Presently, conventional road defect identification techniques may be broadly classified into image processing-based and deep learning-based methodologies[11]. Image processing methodologies commonly utilize techniques including edge detection, Hough transformations, perspective transformations, color thresholding, and multi-modal sensor fusion. Although proficient in uncomplicated situations, their performance diminishes in intricate circumstances[12]. Deep learning techniques, utilizing neural networks for autonomous feature extraction, diminish reliance on manually crafted features. They provide substantial enhancements in detection accuracy, real-time performance, resilience, and generalization abilities, rendering them more appropriate for fault identification across varied road conditions[13]. Deep learning methodologies can be classified into one-stage and two-stage detection methods[14]. Single-stage algorithms accomplish object localization and classification in a singular forward pass, rendering them appropriate for applications requiring high real-time performance[15]. They provide enhanced speeds and reduced computing complexity; yet, they generally exhibit inferior performance

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compared to two-stage algorithms in intricate backdrops and small item identification applications. Two-stage methods (e.g., R-CNN series) markedly improve detection accuracy by initially generating candidate regions and subsequently performing detailed classification, although at the cost of increased computing overhead[16].

In recent years, many academics have focused on creating automated pit detection algorithms, resulting in considerable advancements. Due to the swift progress in computer vision and deep learning, convolutional neural network (CNN)-based detection methods have gained prominence. The YOLO(You Only Look Once) family of models exhibits significant potential for road defect monitoring owing to their end-to-end architecture and real-time detection capabilities[17].

Zhao et al.[18] introduced a feature extraction network utilizing MiWPFE multimodal features and adaptive AMReLU activation functions in traditional machine learning models, markedly improving detection accuracy and generalization in complex environments via frequency-domain entropy correction and attention weight fusion. Ai et al.[19] introduced a support vector machine-based methodology that utilizes multi-scale domain information and pixel intensity to generate probability maps, enhancing certain metrics on the CFD dataset for road problem identification. Saberironaghi et al.[20] introduced an effective U-Net architectural model. The introduction of unique modules to augment feature extraction capabilities resulted in mIoU scores of 77.0% and 83.9% on the Crack500 and DeepCrack datasets, respectively, markedly enhancing the efficacy of automated pavement crack identification.

Despite the notable achievements of current methods in detection accuracy and lightweight design, the robustness and generalization capabilities of models require enhancement when addressing multi-scale defects, inadequate illumination, intense reflective interference, and dynamic occlusions in intricate scenarios[21]. The extensive implementation of edge computing devices in road inspection encounters considerable limitations in computational power, storage capacity, and energy consumption[22].

Researchers have enacted comprehensive enhancements to the YOLO series models to tackle these difficulties. Zhang et al.[23] improved the efficacy of several YOLOv10 architectures, attaining substantial advancements in the recall and localization precision of small objects. Li et al.[24] introduced the lightweight framework RS-YOLO11n, which employs a heterogeneous RHGNetv2 and an occlusion-aware attention module. This method improves detection precision while minimizing model complexity and processing requirements, facilitating real-time implementation on edge devices. These developments have facilitated enhancements in precision and attained lightweight optimization for the detection of road obstacles and defects.

In the context of safe driving, the morphological characteristics of defects and obstacles exhibit significant complexity and diversity. In addition, environmental disturbances and the large parameter scale of detection models make it challenging to simultaneously achieve high recognition accuracy and real-time detection. The baseline model often suffers from inaccurate localization when handling low-contrast obstacles or uneven illumination, which can lead to missed detections. Therefore, this study adopts YOLO11n as the baseline model and proposes a balanced detection framework that simultaneously achieves high accuracy, strong robustness, and lightweight characteristics. The main contributions of this work are as follows:

1) The LSCD_LQE detection head gathers road surface data via shared convolutions and incorporates a Location Quality Evaluator (LQE), facilitating the cooperative optimization of classification and localization. This method greatly enhances the precision of pothole identification while preserving computational economy.
2) The suggested C3k2_RVB module improves multi-scale feature extraction by utilizing re-parameterized deep separable convolutions and channel mixing architectures, alleviating gradient vanishing problems and enhancing defect recognition accuracy in intricate environments. C3k2_REVB integrates an EMA(Efficient Multi-Scale Attention) attention mechanism to improve emphasis on obstacle areas in low-light environments while preserving a lightweight design for resource-limited mobile detection systems.
3) The DySample module accomplishes efficient upsampling through the dynamic generation of adaptive sampling points in conjunction with a grouping technique. This reinstates detail and edge information, rendering it appropriate for real-time multi-scale feature fusion.

Detection module and improvement

YOLO11n network

YOLO is a one-stage object detection system capable of concurrently recognizing object categories and bounding boxes within photos. YOLO11 is the most recent version of the YOLO series launched by Ultralytics, facilitating various visual tasks like as image categorization, object detection, and instance segmentation[25]. The official release includes five scaled variants of the YOLO11 model: YOLO11-n, YOLO11-s, YOLO11-m, YOLO11-l, and YOLO11-x. As the depth and width of the model rise, the number of parameters and computational complexity also escalate. YOLO11 upholds the series' cohesive design principle of conceptualizing object detection as a regression issue, facilitating end-to-end training. This work enhances the YOLO11n model as a foundational reference. The fundamental network architecture consists of three elements: the backbone network, the feature fusion enhancement network (Neck), and the detecting head. Figure 1 depicts the structure of the YOLO11 network.

The Backbone does feature extraction by integrating many layers of convolutional and deconvolutional algorithms. It utilizes residual connections and bottleneck architectures to improve performance while minimizing model size. This part utilizes the C3k2 module as its foundation, which improves computational performance by minimizing redundant parameters while preserving robust feature extraction abilities with a reduced number of parameters. The Neck part, situated between the Backbone and Head, comprises the SPPF (Spatial Pyramid Pooling Fast) module, PAA (Probabilistic Anchor Assignment) module, and PAN (Path Aggregation Network). It improves feature representation by integrating feature maps from various phases of the Backbone. The Head section manages the ultimate detection and classification duties, consisting of a

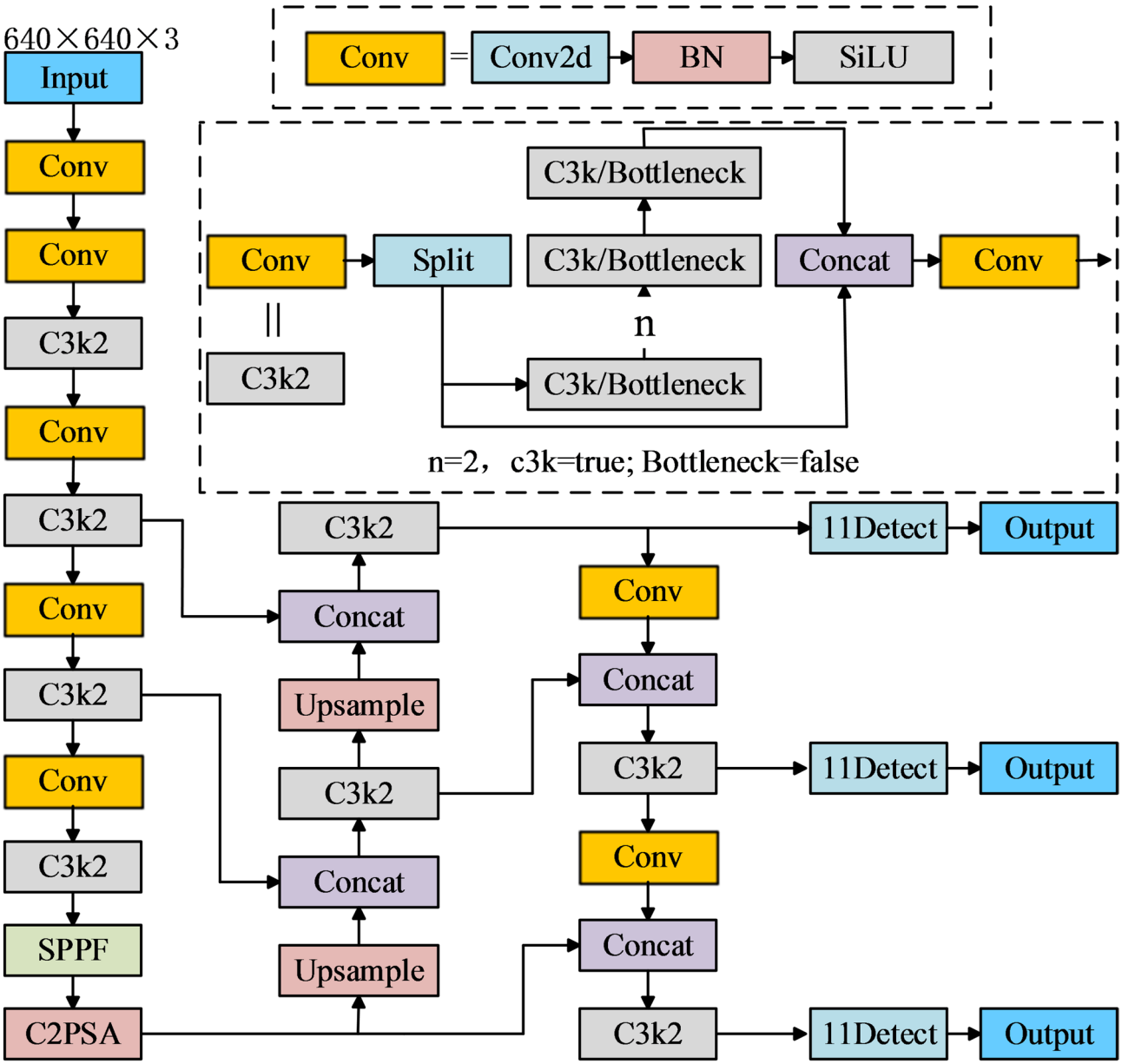


Fig. 1. Diagram of YOLO11 Structure.

detection head and a classification head: The detection head comprises convolutional and deconvolutional layers to forecast bounding box regression values and presence confidence scores for each anchor box, thus producing detection outcomes. The classification head utilizes average pooling to categorize feature maps and produces probability distributions for each category via dimensionality reduction.

RDL-YOLO algorithm

This work presents the RDL-YOLO model, an upgrade of YOLO11n, which principally achieves superior detection accuracy and diminished computing cost by using efficient modules. Initially, in the feature extraction component, we implemented the C3k2 module. This module improves feature extraction by eliminating superfluous parameters and significantly decreasing the model's computational burden. Secondly, the LSCD_LQE shared detection head resolves problems such as indistinct contours and irregular forms of potholes and bumps, enhancing the alignment between detection and localization precision. The Dynamic Up-sampling module (DySample) substitutes conventional up-sampling techniques by dynamically modifying sample sites, thus enhancing the model's efficacy in multi-scale feature fusion. Additionally, by integrating a lightweight Deep Separable Convolution (DWConv) module, we further diminish the model's parameter count and computational complexity, allowing RDL-YOLO to operate efficiently on edge computing devices. Figure 2 depicts the RDL-YOLO architecture.

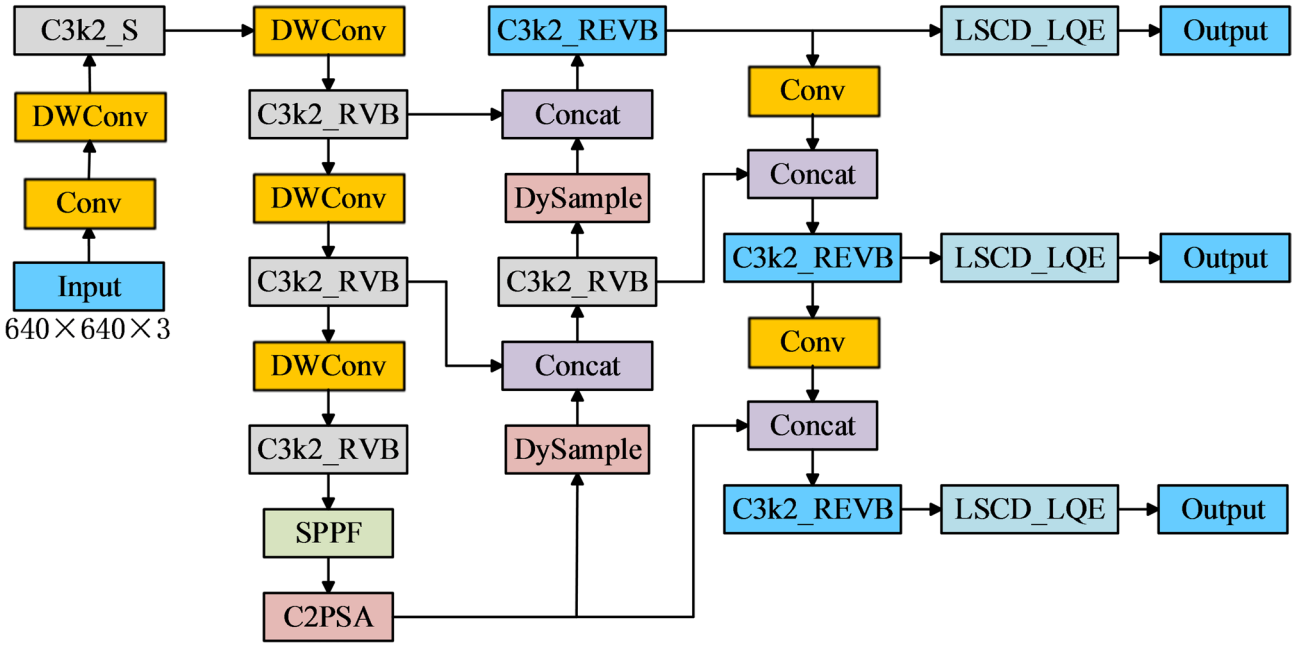


Fig. 2. RDL-YOLO Structure Diagram.

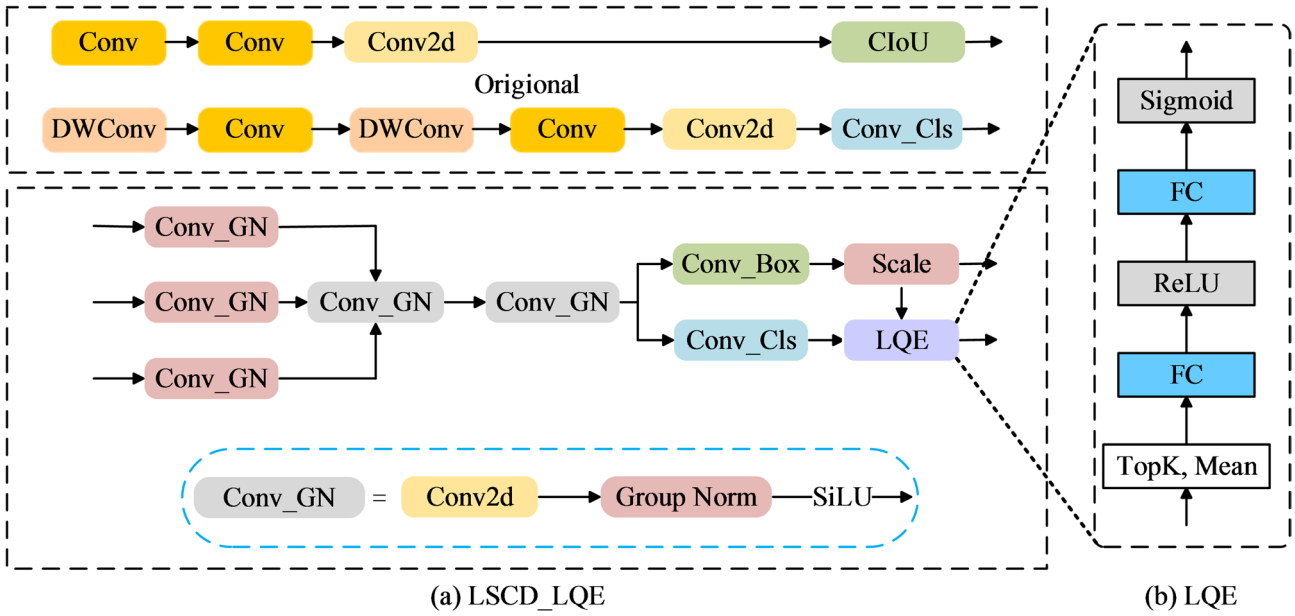


Fig. 3. Structure diagram of LSCD_LQE.

LSCD_LQE detection head

The recognition and localization accuracy of YOLO11 detecting heads are severely mismatched due to problems including blurring edges, irregularly shaped potholes, and projecting impediments. In order to remedy this issue, this study develops an effective shared detection head (LSCD_LQE) that maintains low computing cost while resolving the discrepancy between recognition and localization accuracy. Because of this, it is better suited for putting into practice extremely dependable real-time pothole detection on edge devices. Figure 3a introduces the LSCD detector[26]. Using specialized convolutions, it initially extracts preliminary features from each detection layer's input feature maps. Road surface features are then refined and broken down using shared convolutional layers. Shared convolution normalization effectively reduces model parameters and processing cost while achieving stable representation and feature extraction. Lastly, the network simultaneously predicts the initial categorization scores and the bounding box distribution representation. It increases the adaptability of bounding box regression by adding a learnable scale factor. The final result is the bounding box distribution

concatenated with the calibrated categorization scores. The direct return of layer outputs during training makes calculating loss easier. A dynamic anchor mechanism at inference improves model generalization by adjusting to changing resolution inputs. Precise bounding boxes and sigmoid-transformed classification confidence scores are produced by the final decoding.

In order to greatly decrease false detections, the Location Quality Evaluator (LQE)[27] directly incorporates localization certainty information into classification confidence scores by dynamically analyzing the distribution characteristics of bounding box predictions. Figure 3b displays its structural diagram. By examining the top k high-probability values and their mean within the bounding box distribution, this module efficiently captures localization uncertainty and builds statistical features. After that, it creates a quality adjustment score using a Multi-Layer Perceptron (MLP) and fuses it with the classification score. This method successfully reduces the discrepancy between localization accuracy and prediction confidence. Through the use of the LQE module, the entire procedure optimizes both classification and localization quality, greatly increasing detection accuracy while preserving lightweight features.

C3k2_REVB and C3k2_RVB module

In road defect detection, inadequate brightness, fluctuating lighting conditions, and suboptimal computing architectures hinder the identification of flaws in these intricate situations. The C3k2 architecture of the original YOLO11 has inconsistent recognition accuracy for certain problems and tends to forfeit essential feature information, resulting in gradient vanishing. This paper introduces C3k2_REVB and C3k2_RVB to tackle these issues, as depicted in Fig. 4a. These models reduce recognition mistakes due to lighting fluctuations, and their lightweight architecture is suitable for resource-constrained mobile devices, offering dependable technological support for intelligent detection and avoidance in road infrastructure.

The forward propagation method of the RVB module[28] consists of two primary stages: Token Mixing and Channel Mixing, both tailored for distinct circumstances in road surface fault identification. The principle is depicted in Fig. 4b.

In the Token Mixing phase, the module initially utilizes Reparameterized Deeply Separable Convolution (RepVGGDWConv) for the extraction of local features. This procedure acquires pixel-level neighborhood data using spatial depth convolution, efficiently detecting local texture characteristics of defects like cracks, rutting, and potholes. The reparameterization guarantees a compromise between extensive multi-branch variety in training and structural simplicity in inference. The Channel Mixing step executes feature translation and integration via a bottleneck architecture with residual connections. A 1×1 convolution initially doubles the channel dimension compared to the original input, hence augmenting the model's nonlinear capability. Subsequently succeeded by a GELU activation function to implement a seamless nonlinear transformation; ultimately, a further 1×1 convolution diminishes the channel dimension to the desired output size. When the counts of input and output channels are congruent, the residual connection proficiently alleviates gradient vanishing in deep networks, enabling the integration of low-level detail features with high-level semantic information.

The REVB module enhances RVB by including EMA (Efficient Multi-Scale Attention)[29], as illustrated in Fig. 4c. Compared to standard squeeze-excitation modules, it displays higher sequential modeling capabilities, more effectively addressing identical spatial dependencies in road images and attaining improved performance in identifying road problems under low-light situations. The EMA attention mechanism conducts global context modeling on feature maps. By establishing long-range dependencies among channels, it amplifies the model's attention on pavement problem areas while mitigating background interference from road markings, shadows, water, and other factors.

The enhanced C3k2_RVB module improves multi-scale feature extraction and optimizes the computational architecture for greater efficiency and reduced weight in practical pavement defect detection applications. C3k2_REVB has enhanced robustness in low-light and complex, changeable illumination conditions. This lesson proficiently addresses critical technical issues in detecting road faults in intricate contexts. The lightweight design renders it especially appropriate for use on resource-limited mobile inspection platforms, offering dependable technical assistance for intelligent road infrastructure evaluation and preventative maintenance.

Introducing dynamic upsampling (DySample)

The original neck network architecture replaces the conventional fixed upsampling operation with a Dynamic Upsampling module (DySample). DySample reinterprets feature upsampling from the perspective of point sampling and formulates the reconstruction process as a learnable coordinate transformation. Unlike traditional interpolation methods (e.g., nearest-neighbor interpolation and bilinear interpolation), which rely on predefined sampling kernels, DySample dynamically adjusts sampling positions according to feature distribution, thereby reducing edge blurring and detail ambiguity introduced during spatial scaling.

Given an input feature map $X \in R^{C \times H \times W}$ and an upsampling factor s, the upsampling operation can be formulated as a differentiable sampling function:

$$Y(p) = X(S(p)), \quad p \in \Omega_{sH \times sW} \tag{1}$$

where Y denotes the high-resolution output feature map and $S(p)$ represents the dynamically generated sampling coordinate corresponding to spatial location p.

As illustrated in Fig. 5, DySample constructs a dynamic sampling set S defined as:

$$S = G + O \tag{2}$$

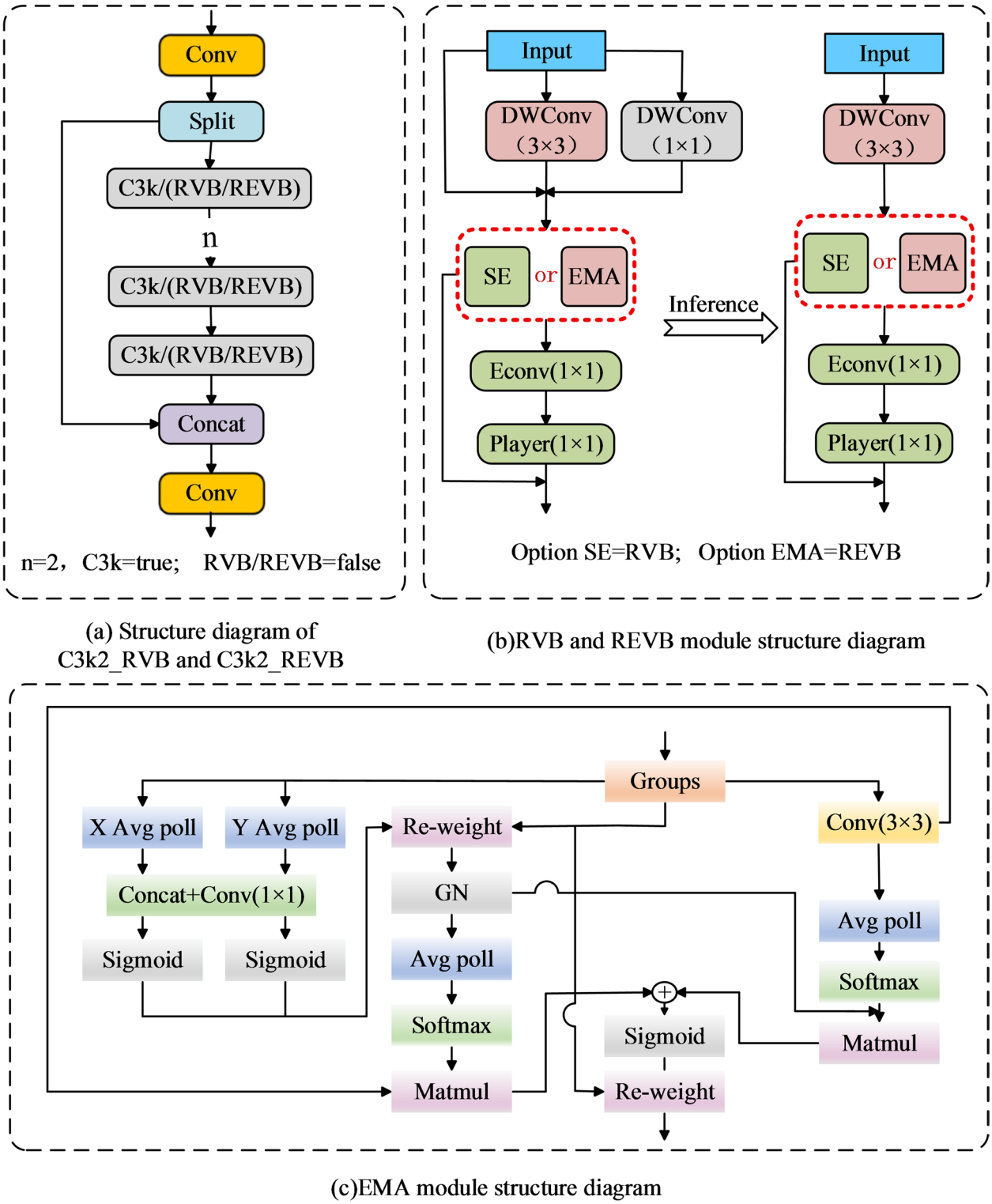


Fig. 4. (**a**) Structure diagram of C3k2_RVB and C3k2_REVB. (**b**) RVB and REVB module. (**c**) EMA module.

where G denotes the initial regular sampling grid and O represents the learnable dynamic offset field. The base grid G determines the default sampling configuration after upsampling, while the offset O adaptively refines these positions according to semantic feature distribution.

The initial sampling grid G is generated according to the upsampling factor s:

$$G = \text{meshgrid}(h, h), \quad h = \frac{1}{s}\left[\frac{-s+1}{2}, \ldots, \frac{s-1}{2}\right] \tag{3}$$

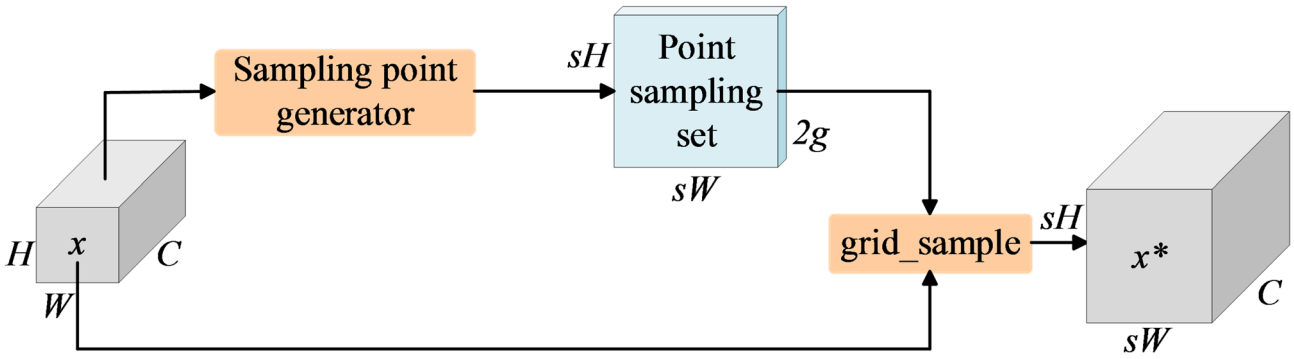


Fig. 5. DySample structure.

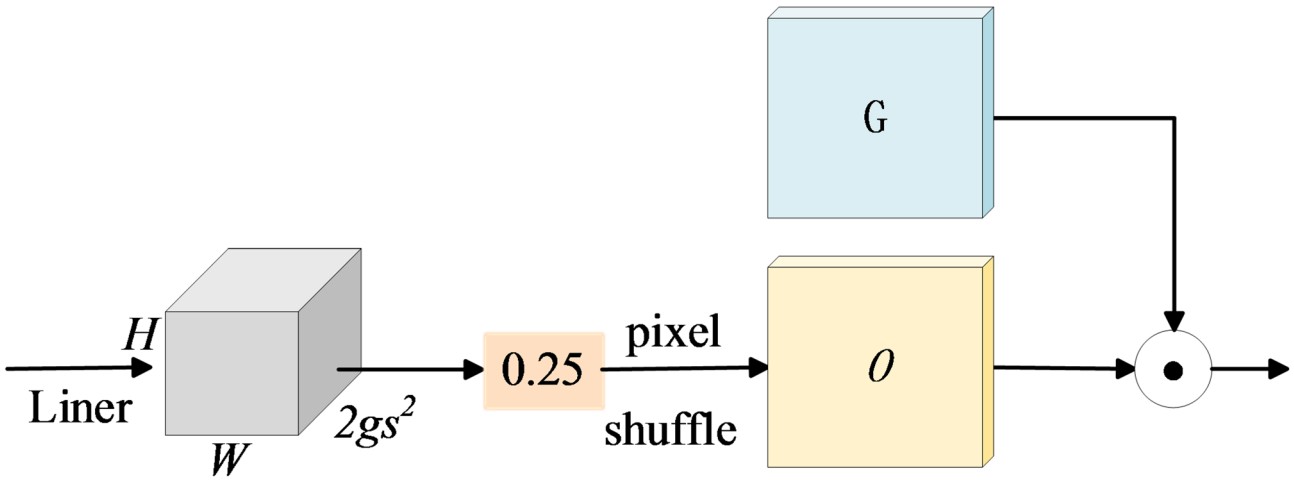


Fig. 6. The generation process of the sampling set S.

This formulation ensures that each pixel in the low-resolution feature map corresponds to s^2 evenly distributed sampling positions in the high-resolution feature space. When the offset term $O=0$, the sampling configuration degenerates to a regular interpolation grid equivalent to bilinear interpolation.

The generation of the dynamic offset O, illustrated in Fig. 6, consists of linear mapping and spatial reorganization. Specifically, a 1×1 convolution layer performs pointwise linear transformation to predict the two-dimensional displacement vector $(\Delta x, \Delta y)$ for each sampling location:

$$O = \mathrm{Conv}_{1\times1}(X) \tag{4}$$

To prevent excessive dispersion or overlap of sampling points and to stabilize the sampling distribution, a range constraint mechanism is introduced:

$$O' = \alpha \cdot \sigma(\mathrm{Conv}_{scope}(X)) \odot O \tag{5}$$

where $\sigma(\cdot)$ denotes the sigmoid function, α is a scaling coefficient, and $\odot$ represents element-wise multiplication. This modulation mechanism effectively restricts the magnitude of offsets and ensures stable spatial transformation. Subsequently, a Pixel Shuffle operation spatially reorganizes the offset tensor, expanding its resolution from *(H, W)* to *(sH, sW)*, so that the predicted offsets align with the target upsampled coordinate grid.

Finally, the high-resolution feature map is obtained through differentiable bilinear resampling:

$$Y(p) = \sum_{q\in\mathcal{N}(S(p))} w(q, S(p))\, X(q) \tag{6}$$

where $\mathcal{N}(S(p))$ denotes the neighboring lattice points around coordinate $S(p)$, and $w(q, S(p))$ represents the bilinear interpolation weights. The resampling process is implemented using the grid_sample operator, enabling end-to-end backpropagation.

Furthermore, to reduce computational overhead, DySample adopts a grouping strategy that divides feature channels into g groups and predicts offsets group-wise. This design reduces parameter redundancy and memory

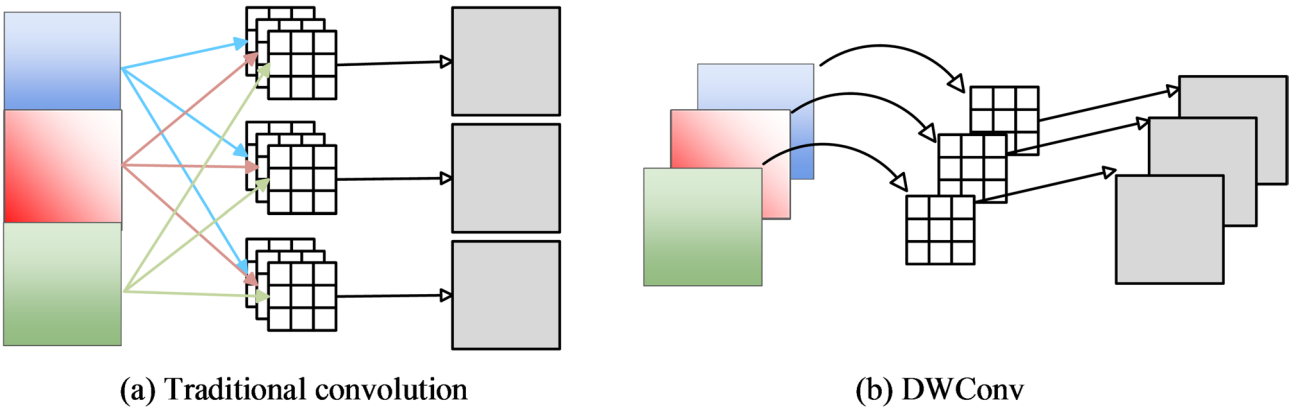


Fig. 7. (**a**) Traditional convolution principle diagram. (**b**) DWConv schematic diagram.

Environment	Configuration
Operating system	Windows10
Python version	Python 3.8
Deep learning framework	PyTorch 2.3.1
CUDA	12.1
CPU	Intel Core i7-12700F
GPU	NVIDIA GeForce RTX 4070
Memory	32 GB

Table 1. Environment Configuration Table.

consumption while maintaining adaptive spatial modeling capability. Compared with conventional full-channel offset modeling, the grouping mechanism improves efficiency without sacrificing detail reconstruction performance, making DySample particularly suitable for lightweight detection frameworks and real-time applications.

Introducing depthwise convolution

Conventional convolutions, depicted in Fig. 7a, necessitate that each channel be convolved with all filters, therefore substantially elevating computational complexity. Conversely, DWConv necessitates that each channel be convolved with a singular filter[31], significantly preserving computational resources. Through the rational compression of the input channel dimension, the model preserves the integrity of feature representation and performance stability while significantly decreasing parameter count and computational complexity, thus realizing an efficient and lightweight network architecture design.

DWConv utilizes a single-channel mode for its convolutional kernels, doing convolution on each input channel separately. This produces an output feature map with an equivalent number of channels as the input feature map. Consequently, the quantity of convolutional kernels corresponds to both the channels of the input feature map and the channels of the output feature map, as depicted in Fig. 7b. To quantitatively delineate the computational attributes of DWConv, consider a convolution kernel size of $k \times k$, an input channel count of ni, an output channel count of no, and output feature map spatial dimensions of $f \times f$. The parameter count is denoted as $k \times ni \times no$, with a computational complexity of $k \times ni \times no \times f^2$. This markedly diminishes both parameter magnitude and computational intricacy in comparison to ordinary convolution. The incorporation of the DWConv module into the YOLOv11 network facilitates successful model lightweighting. This markedly decreases computational expenses while preserving detection precision, exhibiting enhanced suitability in mobile and resource-limited contexts.

Experiments

Experimental environment and dataset

Experimental environment

To guarantee the neutrality and comparability of experimental outcomes, all tests in this work were performed on uniform hardware and software platforms with comparable hyperparameter settings. The testing setup comprised an Intel Core i7-12700F processor, an NVIDIA GeForce RTX 4070 graphics card, 32 GB of RAM, and operated on Windows 10, as detailed in Table 1. The software environment utilized Python 3.8 and CUDA 12.2, employing the PyTorch 2.3.1 deep learning framework for model training and inference.

In the training phase, the input image resolution was configured to 640×640 pixels, utilizing 250 epochs and a batch size of 16. The optimization algorithm utilizes stochastic gradient descent (SGD) with a momentum

coefficient of 0.9, while other hyperparameters are set to their default settings. To maintain uniformity in experimental circumstances, the testing phase employs same input dimensions and parameter configurations as the training phase, ensuring result reproducibility and comparison validity.

Dataset processing and introduction

This research utilizes a dataset including 3755 road surface photos obtained from the Roboflow platform, annotated with the LabelMe tool, to assess the efficacy of the RDL-YOLO model. The dataset categorizes diverse forms of road surface obstacles into a singular class evenly. The selection of road surface defect categories was guided by their representativeness in real-world driving scenarios, visual distinguishability, and the feasibility of acquiring a sufficient number of annotated samples. This strategy ensures a balanced data distribution, thereby facilitating efficient model training and reliable performance evaluation. All images were randomly divided into training, validation, and test sets at a ratio of 7:2:1.

The dataset used for the generalization experiments was sourced from the Kaggle platform and comprises 1,977 images of road obstacles, including multiple categories such as bumps, depressions, and mixed types. All images were randomly divided into training, validation, and test sets with a ratio of 7:2:1. As shown in Fig. 8, the figure presents examples of road obstacles under various natural conditions within the dataset.

As illustrated in Fig. 8, the dataset presents road obstacles captured under diverse natural conditions.

Evaluation indexes

This paper systematically employs four performance metrics—Mean Average Precision (mAP), Precision (P), Recall (R), and F1-Score—as a model evaluation framework to comprehensively, objectively, and intuitively assess and illustrate the improvement effects of the proposed method in this experiment. Furthermore, Three computational efficiency metrics were employed, namely model size, the number of parameters (Params), and frames per second (FPS). The combination of these indicators enables a comprehensive analysis and comparison of the models from multiple perspectives, including detection accuracy, recognition capability, and computational complexity.

Precision quantifies a model's accuracy in predicting positive instances, defined as the ratio of accurately predicted positive instances to the total number of instances projected as positive. Recall denotes a model's capacity to recognize true positives, quantified as the ratio of accurately predicted positive instances to the total number of real positive instances. Their corresponding formulas are:

$$P = \frac{T_P}{T_P + F_P} \tag{7}$$

$$R = \frac{T_P}{T_P + F_N} \tag{8}$$

TP (True Positive): Denotes the quantity of samples that are genuinely positive and accurately classified as positive by the model. FP (False Positive): Denotes the quantity of samples that are genuinely negative yet erroneously classified as positive by the model. FN (False Negative): Denotes the count of samples that are genuinely positive yet erroneously classified as negative by the model.

The F1-Score is the harmonic mean of precision and recall, offering a thorough representation of a model's overall equilibrium and efficacy in prediction accuracy and sample coverage.

$$\mathrm{F1} = \frac{2 \times P \cdot R}{P + R} \tag{9}$$



Fig. 8. Example graph of dataset.

The mean average precision serves as a thorough metric for evaluating a model's detection efficacy. A greater value signifies enhanced model efficacy. The formula for its calculation is:

$$\mathrm{mAP} = \frac{1}{n}\sum_{i=1}^{n} AP_i \tag{10}$$

The mAP50 metric evaluates detection performance at an IoU threshold of 0.5, reflecting the model's ability to correctly identify objects under a relatively lenient localization criterion. In contrast, mAP50–95 computes the mean AP across IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, providing a more comprehensive and stringent assessment of both detection and localization accuracy.

Experimental results and analysis

Ablation experiments

This work conducted eight ablation experiments based on the YOLO11n baseline model to systematically validate the individual contributions and synergistic effects of each enhancement module on model performance. The tests systematically incorporated various enhancement components, such as the C3k2_RVB and C3k2_REVB modules, the LSCD_LQE detection head, the DWConv lightweight convolution, and the DySample dynamic upsampling module. Experiments were performed on the road_pothole dataset, thoroughly assessing metrics like Precision, Recall, F1, mAP50, mAP50-95, GFLOPs, and parameters (Params) to evaluate the balance between detection performance and computing efficiency across model components. The results are encapsulated in Table 2.

The baseline model YOLO11n attained mAP50 and mAP50-95 scores of 90.49% and 54.95%, respectively, using GFLOPs of 6.6 and 2.62 million parameters. The C3k2_RVB and C3k2_REVB modules were initially implemented to assess the effects of residual feature improvement and channel attention fusion on multi-scale feature modeling. The results indicated that mAP50 increased to 91.50%, with both Precision and Recall surpassing 90%, demonstrating that these modules substantially improved semantic consistency and feature discriminability. The LSCD_LQE detection head assesses the significance of local quality estimate in precise feature differentiation. The model exhibits consistent performance with a mAP50 of 91.34% and a mAP50-95 of 54.76%, indicating resilience against small targets and intricate backgrounds.

The independent introduction of the DWConv module examined the structural optimization impacts of lightweight convolutions, decreasing parameters and computing cost to 2.11 M and 4.9 GFLOPs, respectively—approximately 19% and 26% lower than the original model. This attained a mAP50 of 90.06% while markedly improving computing efficiency. Subsequently, the DySample module was implemented to assess the impact of dynamic feature resampling on multi-scale feature fusion. The results showed that a balance was struck between recall and precision, signifying that this method improves feature distribution consistency and spatial perception abilities.

In multi-module combination tests, the integrated design of C3k2_RVB and C3k2_REVB modules with LSCD_LQE sought to collaboratively improve feature extraction and detection heads. This attained a mAP50 of 91.18% while decreasing GFLOPs to 5.1, illustrating complimentary functions in semantic feature propagation and discriminative information extraction. The enhanced integration of the DWConv module attains a balance between precision and lightweight efficiency, while decreasing parameters and GFLOPs to 1.64 M and 3.7, respectively—approximately 37% and 44% lower than the baseline model. Nonetheless, this results in a substantial deterioration of model performance metrics.

By merging the four modules—C3k2_RVB, C3k2_REVB, LSCD_LQE, and DWConv—DySample addresses the constraints of DWConv, resulting in enhanced performance. Precision, Recall, F1, mAP50, and mAP50-95 were 90.82%, 87.21%, 88.98%, 92.25%, and 56.39%, respectively. This setup enhances detection accuracy and real-time performance while preserving low computational complexity.

This ablation design illustrates a sequential validation process from single-structure optimization to multi-module synergistic enhancement. C3k2_REVB and C3k2_RVB fortifies high-level semantic representations,

	Precision%	Recall%	F1%	mAP50%	mAP50-95%	GFLOPs	Size/MB	Params/M
YOLO11n	**91.80**	82.97	87.16	90.49	54.95	6.6	5.3	2.62
①-1	91.03	84.44	87.61	91.22	56.60	5.7	4.8	2.20
①-2	90.82	84.63	87.60	90.96	55.89	5.6	4.8	2.13
①	91.40	85.04	88.11	91.50	**57.16**	5.8	4.8	2.28
②	89.44	86.23	87.81	91.34	57.01	5.6	4.9	2.43
③	87.55	86.33	86.94	90.06	55.92	4.9	4.3	2.11
④	91.03	84.38	87.58	90.26	55.13	6.5	5.3	2.59
①+②	89.46	**87.46**	88.48	91.18	56.63	5.1	4.4	2.12
①+②+③	88.36	85.54	86.93	89.45	53.67	3.7	3.5	**1.64**
①+②+③+④	90.82	87.21	**88.98**	**92.25**	56.39	**3.7**	**3.5**	1.66

Table 2. Ablation experiments. Commentary: ①-1 C3k2_REVB; ①-2 C3k2_RVB①C3k2_REVB and C3k2_RVB; ②LSCD_LQE; ③DWConv; ④DySample.

LSCD_LQE augments the discriminative efficacy of detection heads, DWConv significantly diminishes model complexity, and DySample improves spatial coherence in feature fusion.

Although the final RDL-YOLO exhibits slightly lower precision than YOLO11n, its recall, F1-score, mAP50, and mAP50–95 are all improved. This phenomenon indicates that the proposed enhancements increase the model's sensitivity to diverse road defects, enabling more positive samples to be correctly detected. The marginal reduction in Precision suggests a slight increase in false positives; however, the substantial gain in Recall leads to a higher F1-score, reflecting a more balanced detection performance. In practical road defect detection scenarios, missing defects (false negatives) often incur higher safety risks than limited false alarms. Therefore, the observed trade-off represents a deliberate and reasonable optimization toward robust and reliable defect detection.

The combined implementation of these modules significantly enhances the model's detection performance, attaining a harmonious equilibrium of high accuracy, low complexity, and real-time functionality. This offers a generalizable technological methodology for structural design and performance enhancement in lightweight, multi-scale road flaw detecting systems.

Comparison of different upsampling methods

This paper conducts a comparative experiment to evaluate the influence of various upsampling modules on pavement pothole recognition performance, introducing four methods: classic Upsample, CARAFE, EUCB, and DySample, based on RDL-YOLO. The experimental findings are presented in Table 3.

The findings demonstrate that the DySample module exhibits superior overall performance, attaining Precision, Recall, F1, and mAP50 values of 91.03%, 84.38%, 87.58%, and 90.26%, respectively. DySample enhances recall rate and overall detection capability while preserving excellent detection accuracy. This module augments the model's capacity to identify minute pothole characteristics using an adaptive dynamic sampling approach, hence enhancing resilience and stability in intricate road conditions. Conversely, the conventional Upsample module, while exhibiting a marginally superior Precision of 91.80%, demonstrates a diminished recall rate of merely 82.97%, resulting in potential oversight of detections, which is detrimental to real-time risk mitigation.

The CARAFE module exhibits strong performance in mAP50, achieving 91.64%; however, its recall and F1 scores are marginally inferior to those of DySample, suggesting it excels in feature reconstruction but is less adaptable to irregular pothole targets. The EUCB module has reasonably consistent overall performance; however, its F1 and mAP metrics remain marginally inferior to those of DySample.

In summary, DySample demonstrates superior detection accuracy, recall rate, and stability, markedly enhancing the overall efficacy of pavement pothole recognition algorithms. This study finally identifies DySample as the upsampling module to augment the model's recognition accuracy and vehicle hazard avoidance capability in intricate road settings.

Generalization experiment

This research utilizes publically accessible datasets from Kaggle to rigorously assess the cross-scenario adaptability of the RDL-YOLO model through generalization experiments. This dataset includes 1977 documented incidents of road damage, representing a spectrum of pavement conditions. The data were divided into training, validation, and test sets in a 7:2:1 ratio. Coordinated variable experiments were performed to compare the enhanced algorithm with the original method. Figure 9 illustrates the detection performance of the generalization experiments.

Table 4 illustrates that the enhanced model exhibits greater detection performance across multi-category pothole datasets in comparison to the baseline model. It attains differing levels of enhancement in Recall, F1 score, and mAP50 measures, while preserving similar Precision levels. This demonstrates that RDL-YOLO possesses significant detection stability and accuracy independent of particular data distributions, confirming its exceptional performance and improved generalization ability across many road surface defect detection tasks.

Comparison of the performance of leading algorithms

This paper conducts comparative experiments with several prominent object detection models to systematically validate the extensive performance benefits of the RDL-YOLO algorithm in road surface defect detection tasks. The comparative models encompass the two-stage detection algorithms Faster R-CNN[32] and SSD[33], lightweight single-stage algorithms YOLOv5n, YOLOv8n, and YOLO11n, as well as structurally optimized and feature-enhanced models like Mamba-YOLO-T[34] and hyper-YOLO[35]. Table 5 presents the performance metrics for each model, encompassing accuracy, recall, computational complexity, and model size.

Experimental findings indicate that RDL-YOLO attains remarkable detection efficacy while preserving notably low model complexity. Its mAP50 attains 92.25% and F1-Score 88.98%, surpassing all comparative models. Simultaneously, RDL-YOLO's parameter size is only 3.5 MB, with computational demands of 3.7GFLOPs,

Module	Precision%	Recall%	F1%	mAP50%
Upsample	**91.80**	82.97	87.16	90.49
CARAFE	90.77	81.93	86.12	91.64
EUCB	90.69	84.19	87.31	91.15
Dysample	91.03	**84.38**	**87.58**	90.26

Table 3. Comparison of different upsampling methods.



Fig. 9. Generalization experiment detection effect diagram.

Module	Precision%	Recall%	F1%	mAP50%
YOLO12n	77.69	69.58	73.42	76.22
YOLO11n	**80.77**	68.38	74.06	76.73
YOLOv8n	79.37	71.75	75.38	76.62
YOLOv5n	77.77	70.60	74.02	76.44
RDL-YOLO(our)	80.69	**75.64**	**78.08**	**79.22**

Table 4. Generalization experiment comparison table.

	Precision%	Recall%	F1%	mAP50%	mAP50-95%	FLOPs/G	Model Size	Params/M	FPS
Faster-R-CNN	79.61	69.22	74.05	80.77	48.34	370.2	108.2 MB	136.8	23
SSD	73.34	67.38	70.23	73.31	44.89	128.9	49.1 MB	32.3	59
Mamba-YOLO-T	83.24	75.60	79.23	82.69	50.22	13.6	12.3 MB	6.0	184
hyper-YOLO	88.42	85.69	87.04	91.03	55.47	9.5	7.3 MB	3.62	209
RT_DETR	90.05	84.67	87.28	88.93	52.66	42.6	32.5 MB	16.76	138
YOLO12n	88.41	83.67	85.97	89.33	53.82	6.7	5.3 MB	2.60	307
YOLOv8n	89.76	84.94	87.24	90.24	54.92	8.9	5.4 MB	2.69	234
YOLOv5n	88.47	85.04	86.72	89.82	54.27	7.1	5.0 MB	2.51	295
YOLO11n	**91.80**	82.97	87.16	90.49	54.95	6.6	5.3 MB	2.62	363
RDL-YOLO (our)	90.82	**87.21**	**88.98**	**92.25**	**56.39**	**3.7**	**3.5 MB**	**1.66**	**543**

Table 5. Comparative experiment.

considerably less than that of most prevalent models. In comparison to the baseline model YOLO11n, RDL-YOLO demonstrates a 1.35% enhancement in mAP50 and a 1.3% rise in F1-Score, concurrently decreasing model parameters and computing demands by about 25% and 44%, respectively, so showcasing substantial benefits in lightweight architecture and efficiency.

A thorough investigation reveals that while two-stage detection models (e.g., Faster R-CNN and SSD) exhibit robust feature extraction skills, their intricate detection mechanisms and reduced inference speeds render them unsuitable for real-time detection needs. Transformer-based detectors, exemplified as Mamba-YOLO-T, demonstrate promise in feature representation; nonetheless, they are constrained by issues of model stability and accuracy. In contrast, RDL-YOLO attains an advantageous equilibrium among detection accuracy, computational efficiency, and model size. It precisely detects various defect types—such as cracks and potholes—in intricate road settings, offering dependable visual perception assistance for real-time hazard evasion by autonomous vehicles and road safety surveillance. Detection outcomes are depicted in Figs. 10 and 11.

Conclusions

Experimental findings indicate that the RDL-YOLO model attains substantial enhancements in the detection of road surface defects. The integration of C3k2, LSCD_LQE, DWConv, and DySample modules enhanced the model's mAP50 from 90.49% in YOLO11n to 92.25%, with recall and F1 score achieving 87.21% and 88.98%,

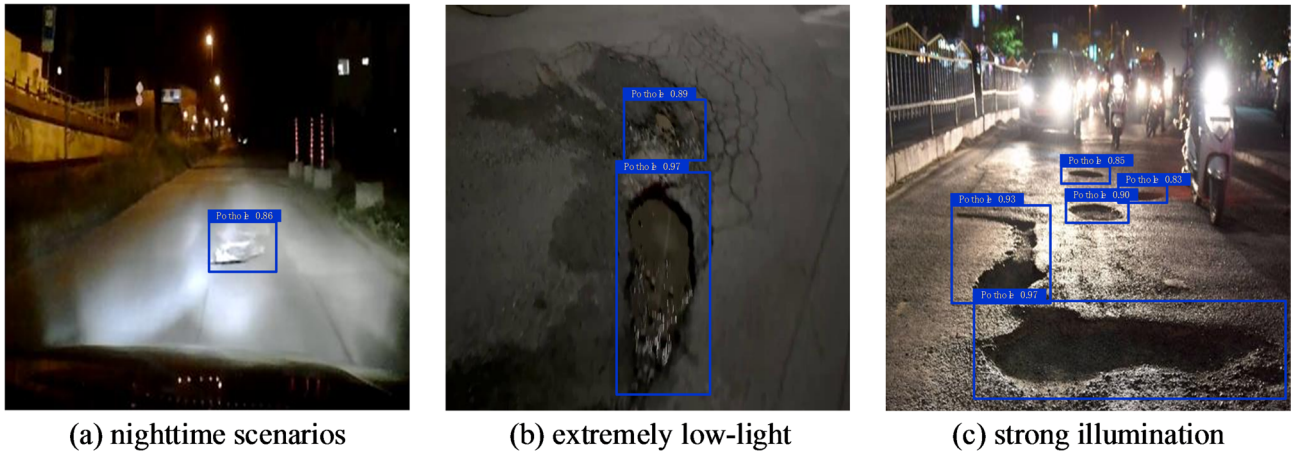


Fig. 10. RDL-YOLO detection performance in different scenarios.

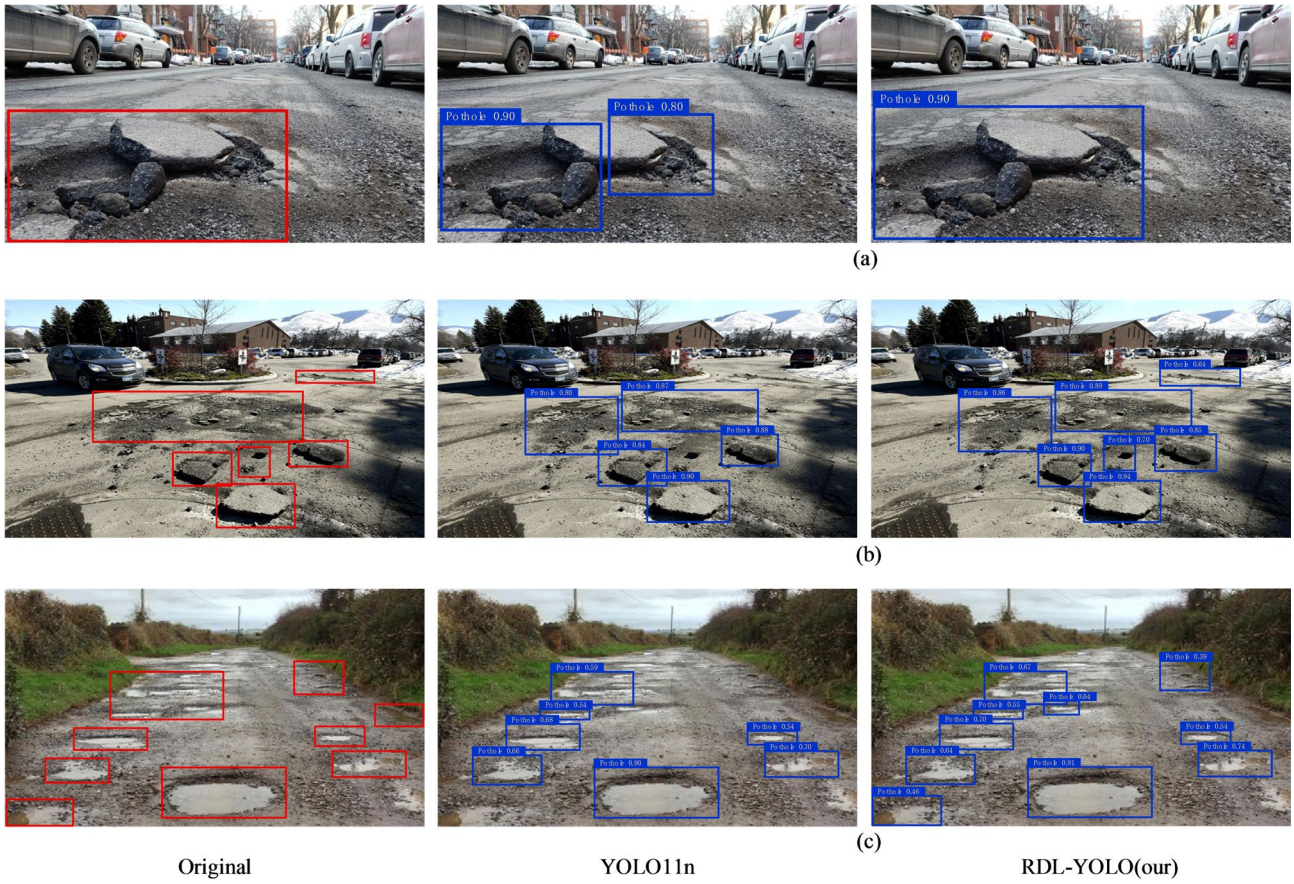


Fig. 11. Detection results of RDL-YOLO and YOLO11n under different environments. (**a**) Cities. (**b**) Towns. (**c**) Rural areas.

respectively. RDL-YOLO achieves a computational efficiency with GFLOPs reduced to 3.7 and model parameters to 1.66 M, reflecting decreases of 44% and 25% relative to YOLO11n. Moreover, generalization experiments confirm the model's stability and exceptional accuracy across several road defect datasets, illustrating its adaptability across different scenarios.

The RDL-YOLO model markedly improves the precision and real-time efficacy of road defect identification via a succession of new modules, exhibiting notable robustness, especially in intricate situations. In comparison to current YOLO series models and other sophisticated algorithms, RDL-YOLO demonstrates distinct improvements in precision, computational efficiency, and model dimensions. It enhances identification accuracy

for small objects while preserving low computational complexity, rendering it appropriate for pavement fault detection in intelligent transportation systems. The RDL-YOLO model can be structurally adjusted to enhance real-time detection applications, including autonomous driving and intelligent vehicle hazard avoidance. However, the practical applicability of this method in rugged terrains such as the Gobi Desert, barren lands, and karst landscapes remains to be improved, and its lateral applicability is limited. Future efforts will focus on scenario-oriented refinements tailored to specific regions.

Data availability

The data that support the findings of this study are openly available at the following URL: https://universe.roboflow.com/project-5iscr/-sd68a. https://www.kaggle.com/datasets/anggadwisunarto/potholes-detection-yolov8.

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Author contributions

X.Z.: methodology, software, validation, Writing -original draft, Writing—review & editing; H.J.: data curation, validation; N.C.:data curation, validation; S.D.: data curation, supervision; Z.F.: project administration, supervision. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no conflicts of interest.

Additional information

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